WEEK 6 (Mon. / Thu.)

BEFORE YOU START — WHAT TO COLLECT

Inorganic Experiment 2, Session II — purification and six characterizations

THIS SESSION IS A TIMETABLING PROBLEM, NOT A CHEMISTRY PROBLEM

One purification feeds six measurements, four of which are on shared instruments. **Start the sublimation first and run everything else while it proceeds** — each cycle takes time in which you are otherwise idle.

A workable order: weigh the covers, load the dish, start heating at 100 °C, and while cycle 1 runs take the crude ferrocene image on the stereomicroscope. Between cycles, weigh. Once enough purified material is in hand, grind for the magnetic balance, then weigh out the UV–Vis sample, then make up the NMR and ESI-MS tubes last so they go in before the submission cut-off.

A. Glassware and apparatus

☐ Glass petri dish **base** (smaller)
☐ Glass petri dish **covers** (larger) × **2**
☐ 400 mL beaker — for the ice water
☐ Hotplate
☐ Agate mortar and pestle
☐ Susceptibility sample tube + ruler
☐ **25 mL volumetric flask** — not a measuring cylinder
☐ Quartz UV cuvette + lint-free wipes
☐ Pasteur pipettes for the dilutions
☐ Small beakers / vials for the dilution series
☐ NMR tube + cap + label
☐ Sample tube for ESI-MS + label
☐ Transparent sample plastic bag or vial for the product
☐ Spatula, weighing boats, watch glass

B. Reagents and consumables

☐ **Your crude ferrocene** — from Week 5
☐ Ice, for the cover beaker — take plenty, it melts
☐ Methanol for the UV–Vis solution
☐ $CDCl_3$, **0.6 mL**
☐ KBr, only if you choose the KBr disc route
☐ Acetone and methanol for washing glassware

C. Shared equipment — four instruments, one afternoon

☐ **Analytical balance** — needed repeatedly, every cycle
☐ **Stereomicroscope** at 14× + camera
☐ **Magnetic susceptibility balance** (Sherwood MSB AUTO)
☐ **UV–Vis spectrometer**
☐ **ATR-IR**, or the KBr die and press
☐ NMR and ESI-MS submission points, and the cut-off times

D. Personal and paperwork

☐ Own lab coat and safety goggles
☐ Full-length trousers, covered shoes, high-cut socks
☐ **Phone, charged** — five MSB photographs plus two microscope images
☐ This bench sheet and the datasheet, printed
☐ **Week 5 microscope image**, for the before-and-after comparison
☐ Calculator, permanent marker, S/N numbers

THREE THINGS THAT ARE EASY TO LOSE MARKS ON

Weigh both covers empty before you load anything. The purified yield is the sum of cover-plus-ferrocene minus cover-empty over every cycle. An unweighed cover is an uncountable cycle.

The UV–Vis mass must be accurate to four decimals. ε is computed from it, so a mass recorded as "about 38 mg" makes the extinction coefficient unreportable. Weigh 35–40 mg on the analytical balance and write down every digit.

Photograph the MSB screen *before* the last digit. The display advances the instant you press it. This is five separate photographs, and the manual marks them as assessed.

요약. 이번 세션의 난점은 화학이 아니라 일정이다. 정제 하나에 측정 여섯 개가 붙어 있고 그중 넷이 공용 장비다. 승화를 가장 먼저 걸어 두고 사이클이 도는 동안 나머지를 처리한다. 뚜껑 칭량 → 시료 장입 → 100 °C 가열 시작 → 1차 사이클 도는 동안 조생성물 현미경 촬영 → 사이클 사이마다 칭량 → 충분히 모이면 분쇄 후 자기감수율 → UV-Vis용 칭량 → NMR과 ESI-MS 시료는 제출 마감 전에 마지막으로 만드는 순서가 무난하다. 감점으로 이어지기 쉬운 셋을 짚어 둔다. 뚜껑은 반드시 비운 상태로 먼저 칭량한다. UV-Vis 시료는 소수점 넷째 자리까지 정확히 재야 ε을 낼 수 있다. 자기감수율 저울은 마지막 자릿수를 누르기 전에 찍어야 하며, 사진이 다섯 장 필요하고 평가 대상이다.

WEEK 6 (Mon. / Thu.)

INORGANIC EXPERIMENT 2 — SESSION II

Purification and Characterization of Ferrocene

Venue	General & Synthetic Chemistry Lab, Block S5 Level-1	**Instrument / technique**	Stereomicroscope, MSB, IR, UV–Vis, ESI-MS, ^{1}H NMR
Time	10:00 a.m. – 4:00 p.m. (Mon. or Thu.)	**Source**	Manual v0724 + lecture notes

(a) Purification by sublimation

1. **Weigh both larger petri dish covers empty** on the analytical balance and record each.
2. Spread the crude ferrocene on the smaller petri dish base; cover with a weighed cover.
3. Raise the hotplate *slowly*; **100 °C for cycle 1**, and never past **150 °C**. Place the 400 mL beaker of ice water on the cover.
4. Take the cover off the heat, let it cool, recover the sublimed ferrocene. Repeat until nothing further sublimes.
5. Weigh each loaded cover; transfer the product to a clean transparent bag or vial.

WHAT TO EXPECT FROM EACH CYCLE — SEE APPENDIX G

Cycle 1 mostly moves **water**: the crude is still damp from last week's filtration, so expect a wet, pale cover and little orange. Raise the temperature from cycle 2 or 3 and the orange flakes appear. Non-volatile impurities stay on the *base*, which darkens as the cycles proceed — that residue is the evidence that purification happened.
Let the cover cool before handling, invert it over the bag and not the bench, and re-weigh the cover empty before reusing it. Only two are issued.

(b) Characterization — six measurements

1. **Stereomicroscopy.** 14×, one image, matriculation number typed on it. Compare against the Week 5 crude image; this pair is the qualitative evidence for purification.
2. **Magnetic susceptibility.** Grind **0.09–0.12 g** in the agate mortar. Both χ_v and χ_g, five photographs. *Read Appendix H before computing a moment; Appendix I shows why the answer must be zero.*
3. **UV–Vis. 35–40 mg** weighed to four decimals, in methanol, **25 mL volumetric flask**. Record 300–700 nm; dilute until $A = 0.1$–1.0. Determine ε for each peak; the calculation must be handwritten on the spectra.
4. **IR.** Solid spectrum 500–4000 cm^{-1}, ATR or KBr. Record which. *Appendix C.*
5. **^{1}H NMR. ∼1 mg** into a tube containing **0.6 mL $CDCl_3$**. One sharp singlet near δ 4.15 for all ten protons is the result — the ten C–H are equivalent because both rings rotate freely and all five carbons of each bind iron equally.
6. **ESI-MS. 1–2 mg** into a sample tube. *Appendix B.*

THE MSB PROCEDURE — FIVE PHOTOGRAPHS, ALL ASSESSED

Switch on and wait for a zero reading (−0.000E−5Z) [**photo 1**]. Weigh an empty tube, place it on the detector, read, press TARE, confirm zero. Remove it, introduce the ground sample, measure the packed length (**15.0–25.0 mm**) and mass (**0.0010–0.9999 g**). Press LENGTH and enter three digits [**photo 2, before the last digit**]. Press WEIGHT and enter [**photo 3, before the last digit**]. Press MAG SUS and record χ_v [**photo 4**] and χ_g [**photo 5**]. The display advances the instant the last digit is entered.

Before you leave: ☐ Purified ferrocene labelled ☐ Microscope image ☐ Five MSB photos ☐ IR and UV–Vis spectra ☐ ESI-MS and NMR submitted ☐ Datasheet signed

6주차 (월 / 목)

무기 실험 2 — 세션 II

페로센의 정제와 특성분석

장소	General & Synthetic Chemistry Lab, Block S5 Level-1	**장비**	실체현미경, MSB, IR, UV--Vis, ESI-MS, ^{1}H NMR
시간	오전 10시 – 오후 4시 (월 또는 목)	**출처**	매뉴얼 v0724 + 강의노트

(a) 승화에 의한 정제

1. **큰 페트리 접시 뚜껑 두 개를 빈 상태로 먼저 칭량**하고 각각 기록한다.
2. 작은 페트리 접시 바닥에 조 페로센을 펴고, 칭량한 뚜껑을 덮는다.
3. 핫플레이트 온도를 천천히 올린다. **1차 사이클은 100 °C**로 하고 **150 °C**를 넘기지 않는다. 얼음물을 담은 400 mL 비커를 뚜껑 위에 올린다.
4. 뚜껑을 열에서 내려 식힌 뒤 승화된 페로센을 회수한다. 더 이상 승화되지 않을 때까지 반복한다.
5. 페로센이 붙은 뚜껑을 칭량하고, 생성물을 깨끗한 투명 비닐백이나 바이알에 옮긴다.

사이클별로 무엇이 일어나는가 — 부록 G 참조

1차 사이클은 사실상 **물**을 옮기는 과정이다. 지난주 감압 여과로 얻은 조생성물이 아직 젖어 있어 뚜껑이 축축하고 주황색이 거의 없어도 정상이다. 2–3차부터 온도를 올리면 주황색 박편이 나타난다. 비휘발성 불순물은 바닥에 남아 사이클이 진행될수록 짙어지는데, 그 잔사가 곧 정제가 일어났다는 증거다.
뚜껑은 식힌 뒤에 다루고, 실험대가 아니라 비닐백 위에서 뒤집는다. 재사용하기 전에는 비운 상태로 다시 칭량한다. 뚜껑은 두 개뿐이다.

(b) 특성분석 — 여섯 가지 측정

1. **실체현미경.** 14배, 이미지 한 장, 학번을 이미지에 넣는다. 5주차 조생성물 이미지와 비교하며, 이 한 쌍이 정제의 정성적 증거가 된다.
2. **자기감수율.** 마노 유발에서 **0.09–0.12 g**을 곱게 간다. χ_v와 χ_g 모두, 사진 다섯 장. 모멘트를 계산하기 전에 부록 H를 읽을 것. 답이 0이어야 하는 이유는 부록 I에 있다.
3. **UV--Vis. 35–40 mg**을 소수점 넷째 자리까지 정확히 재어 메탄올에 녹이고 **25 mL 부피플라스크**를 쓴다. 300–700 nm를 측정하고 $A = 0.1$–1.0이 되도록 희석한다. 각 피크의 ε을 구하며, 계산은 스펙트럼 위에 손으로 쓴다.
4. **IR.** 500–4000 cm^{-1} 고체 스펙트럼을 ATR 또는 KBr로 측정하고 어느 쪽인지 기록한다. 부록 C.
5. **^{1}H NMR.** $CDCl_3$ **0.6 mL**가 든 튜브에 **약 1 mg**을 넣는다. 결과는 δ 4.15 부근의 날카로운 단일선 하나이며 10개 수소가 모두 여기에 해당한다. 두 고리가 자유롭게 회전하고 각 고리의 탄소 다섯 개가 모두 동등하게 철과 결합하기 때문이다.
6. **ESI-MS.** 샘플 튜브에 **1–2 mg**을 넣는다. 부록 B.

MSB 조작 — 사진 다섯 장, 모두 평가 대상

전원을 켜고 0 표시(−0.000E−5Z)를 기다린다 **[사진 1]**. 빈 튜브를 칭량해 검출부에 올리고 값을 읽은 뒤 Tare를 눌러 0을 확인한다. 튜브를 빼고 분쇄한 시료를 채운 뒤 충전 길이(**15.0–25.0 mm**)와 질량(**0.0010–0.9999 g**)을 측정한다. Length를 누르고 세 자리를 입력한다 **[사진 2, 마지막 자리 누르기 전]**. Weight를 누르고 입력한다 **[사진 3, 마지막 자리 전]**. Mag Sus를 눌러 χ_v **[사진 4]**와 χ_g **[사진 5]**를 기록한다. 마지막 자리를 누르는 순간 화면이 넘어간다.

퇴실 전 확인: ☐ 정제 페로센 라벨링 ☐ 현미경 이미지 ☐ MSB 사진 5장 ☐ IR·UV--Vis 스펙트럼 ☐ ESI-MS·NMR 제출 ☐ 데이터시트 서명

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 6 (1 of 2)

Session II: Sublimation Record

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Starting material

Mass of crude ferrocene from Week 5		Hotplate setting, cycle 1	
Cover A, empty (g)		Cover B, empty (g)	

2. Sublimation cycles

Cycle	Set / °C	Cover	Cover + Fc (g)	Fc (g)	Appearance of cover / base
1		A / B			
2		A / B			
3		A / B			
4		A / B			
			Total purified ferrocene (g)		

3. Colour checkpoints

Stage	Expected	Observed / time	✓
Crude, before heating	orange to orange-brown powder		
Cover after cycle 1	**wet and pale** — mostly condensed water, little orange		
Cover after cycles 2–3	orange flakes or plates		
Petri dish **base**, later cycles	darkens; non-volatile residue left behind		
Purified product	bright orange crystalline flakes		
After grinding for the MSB	light orange free-flowing powder		
Methanol solution for UV	clear yellow-orange		

Expected is a checkpoint, not an answer. A pale cover after cycle 1 is the normal result, not a failure; record what you see and the time.

To be completed at the end of the session		
Student signature	GTA name	GTA signature

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 6 (2 of 2)

Session II: Characterization Record

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

4. Magnetic susceptibility — Sherwood MSB AUTO

Zero reading		Packed length L (mm)	
Empty tube mass (g)		Sample mass m (g)	
Volume susceptibility χ_v		Mass susceptibility χ_g	
Room temperature (K)		Sign of χ_g ☐ − ☐ 0 ☐ +	

Photographs: ☐ zero ☐ before last digit of length ☐ before last digit of weight ☐ χ_v ☐ χ_g **A reading of exactly zero is a resolution limit, not a moment — Appendix H.**

5. UV–Vis

Mass weighed (g, four decimals)		Path length (cm)	1.00
Flask volume (mL) / solvent	25, MeOH	Concentration c (M)	
Dilution factor, if any		λ_1 (nm) / A_1	
Expected λ: ∼440 and ∼325 nm		λ_2 (nm) / A_2	

6. IR, NMR, ESI-MS and imaging

IR method ☐ ATR ☐ KBr		Mass for ^{1}H NMR (∼1 mg)	
Microscope image of purified ☐		Mass for ESI-MS (1–2 mg)	
S/N no. for NMR		S/N no. for ESI-MS	

7. Yield and observations

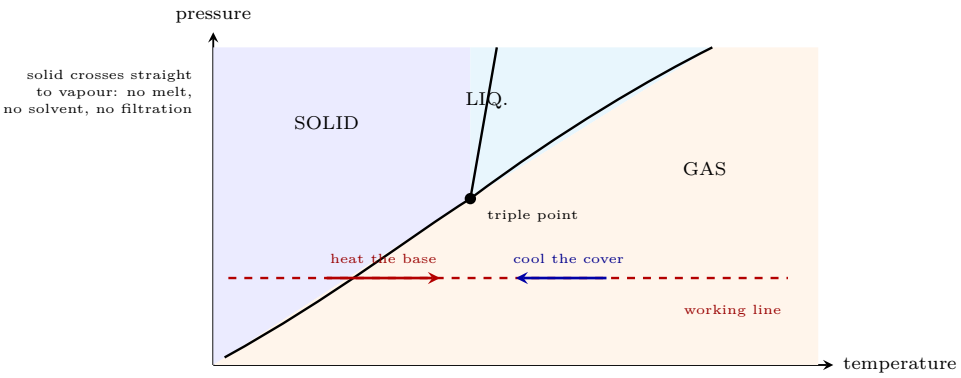
Purified mass / crude mass, and the % recovered	Crude vs purified under the microscope — size, habit, transparency

To be completed at the end of the session		
Student signature	GTA name	GTA signature

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

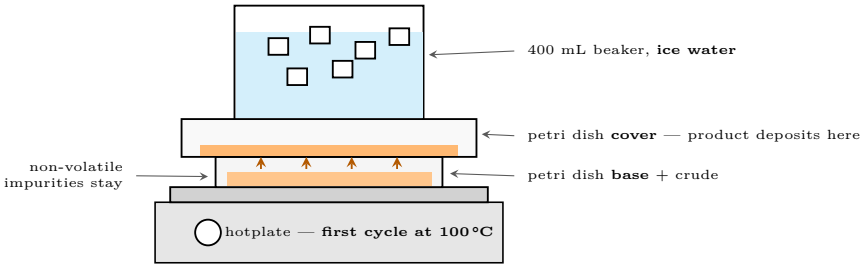
Appendix G — Purification by Sublimation

G1. Where sublimation sits on the phase diagram



Below the triple point a solid has no liquid field to pass through: heating drives it straight to vapour, and cooling brings it straight back to a solid. That is the whole method. In this experiment the dish is open to the room rather than evacuated, so strictly what you exploit is that **ferrocene has a useful vapour pressure well below its melting point of 172.5 °C** while the impurities beside it have effectively none. Heating the base raises that vapour pressure; the ice-water beaker on the cover creates the cold surface the vapour needs to land on. Remove either half of the gradient and nothing transports.

G2. The apparatus, and what ends up where



Cycle 1 is mostly water. The crude came off a suction filter last week and is still damp, so the first pass deposits condensed water on the cover along with very little ferrocene. That is why the first cycle is run at **100 °C** and why you should not be alarmed by a wet, pale cover. Raise the temperature only from the second or third cycle, and never past **150 °C** — above the melting point you get a melt rather than a sublimate, and the transport stops.

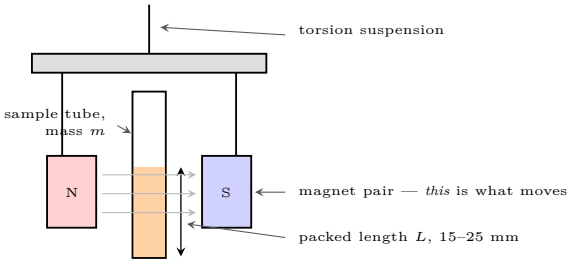
Weigh every cover empty, first. The yield is the difference between cover-plus-ferrocene and cover-empty, summed over all cycles, so a cover you forgot to weigh is a cycle you cannot count. Only two covers are issued: clean, dry and re-weigh them between cycles.

Handle the loaded cover carefully. Let it cool before touching it, and invert it over the collection bag rather than over the bench. Product on the bench is product lost, and unlike a filtration there is no mother liquor to recover it from.

요약. 승화는 고체가 액체를 거치지 않고 바로 기체가 되었다가 다시 고체로 붙는 현상을 이용한다. 이 실험은 감압이 아니라 개방계이므로, 정확히는 페로센이 녹는점 172.5 °C보다 한참 낮은 온도에서도 쓸 만한 증기압을 갖는 반면 불순물은 그렇지 않다는 점을 이용하는 것이다. 바닥을 가열해 증기압을 올리고 뚜껑을 얼음물로 식혀 증기가 앉을 찬 면을 만든다. 둘 중 하나만 빠져도 이동이 일어나지 않는다. 1차 사이클은 사실상 물을 날리는 과정이다. 지난주 감압 여과로 얻은 조생성물이 아직 젖어 있기 때문이며, 그래서 100 °C에서 시작하고 뚜껑이 축축하고 색이 옅어도 정상이다. 2–3차부터 온도를 올리되 150 °C를 넘기지 않는다. 녹는점을 넘으면 승화가 아니라 용융이 되어 이동이 멈춘다. 뚜껑은 반드시 비운 상태로 먼저 칭량한다. 수율은 사이클마다 뚜껑 무게 차의 합이므로, 칭량하지 않은 뚜껑은 세지 못하는 사이클이다.

Appendix H — Magnetic Susceptibility, and the Trap in the Arithmetic

H1. What the balance measures



A paramagnetic sample pulls the magnets *towards* it; a diamagnetic one pushes them away.

The Sherwood balance inverts the classical Gouy experiment. Instead of weighing a sample hung in a field, it suspends the *magnets* and measures the reaction force when the sample is introduced. The instrument converts that deflection, together with the length and mass you type in, into two numbers: the **volume susceptibility** χ_v, which is dimensionless, and the **mass susceptibility** χ_g in $\mathrm{cm^3\ g^{-1}}$. They are related through the packing density, $\chi_v = \chi_g \rho$, which is why the packed length matters as much as the mass and why the powder must be ground evenly in the agate mortar.

H2. From χ_g to a moment

$$\chi_M = \chi_g M \qquad \chi_M^{\mathrm{corr}} = \chi_M - \chi_M^{\mathrm{dia}} \qquad \mu_{\mathrm{eff}} = 2.828\sqrt{\chi_M^{\mathrm{corr}} T} \qquad \mu_{\mathrm{eff}} = \sqrt{n(n+2)}$$

$M = 186.04\ \mathrm{g\ mol^{-1}}$, T in K, everything in cgs. The diamagnetic correction χ_M^{dia} is the contribution of all the paired electrons, estimated from Pascal's constants; for ferrocene it is about $-1.3 \times 10^{-4}\ \mathrm{cm^3\ mol^{-1}}$. It is *negative*, so subtracting it **adds** a positive quantity to χ_M.

THE TRAP — READ THIS BEFORE YOU WRITE A MOMENT DOWN

Ferrocene is an 18-electron, low-spin d^6 complex. Every electron is paired, so $n = 0$ and μ_{eff} should be **zero**. A correctly behaving sample gives a small *negative* χ_M, roughly equal to the diamagnetic correction itself, so that χ_M^{corr} comes out at zero.

Now look at what happens if the balance simply reads $\chi_g = 0$, which it very often will because the true value sits at the edge of its resolution. Then

$$\chi_M^{\mathrm{corr}} = 0 - (-1.3 \times 10^{-4}) = +1.3 \times 10^{-4} \qquad \Rightarrow \qquad \mu_{\mathrm{eff}} \approx 0.6\text{–}0.8\ \mathrm{B.M.}$$

and a moment appears out of nowhere. **It is an artefact of subtracting a correction the measurement was never sensitive enough to detect**, not evidence of unpaired spin. Reporting 0.78 B.M. as a measured moment is the single most common error in this experiment.

Write it instead as an **upper bound**: the sample is diamagnetic within the resolution of the balance, consistent with $n = 0$. If you want to be quantitative, state the instrument's resolution and quote $n < 0.3$ rather than $n = 0.27$.

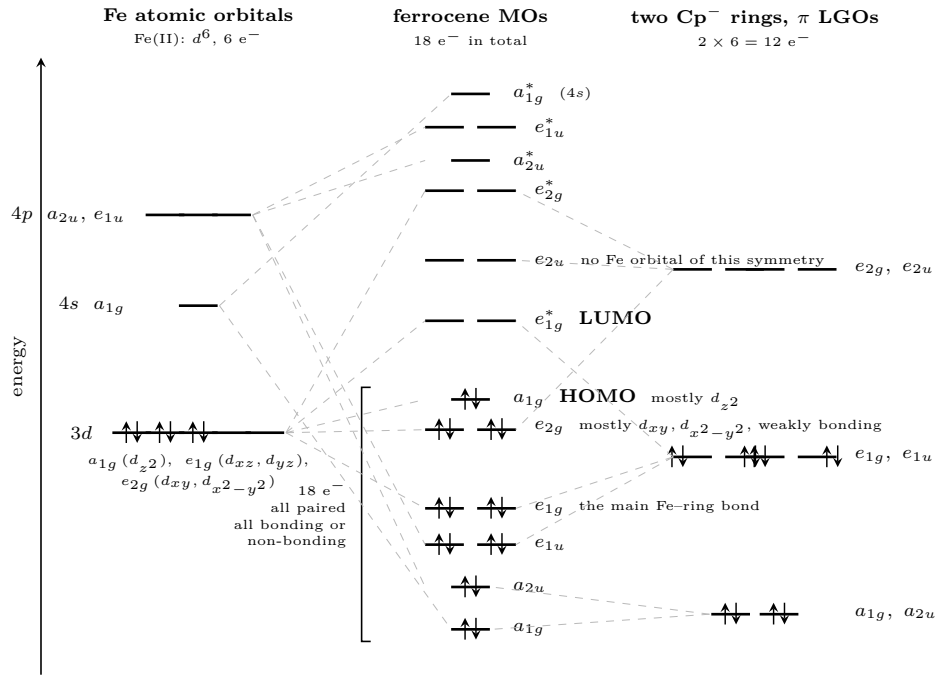
H3. Practical points that decide whether the number means anything

1. **Grind properly.** 0.09–0.12 g in the agate mortar until it is a free-flowing powder. Uneven packing changes ρ, and ρ sits directly in χ_v.
2. **Tap the tube down** so the column is level and continuous. A void in the middle is a length you measure but a mass that is not there.
3. **Length 15.0–25.0 mm, mass 0.0010–0.9999 g.** Outside those windows the instrument's calibration does not apply and the reading is meaningless rather than merely imprecise.
4. **Photograph five moments**: the zero reading, the screen before the last digit of the length, the screen before the last digit of the weight, χ_v, and χ_g. The display advances the instant the last digit is entered, so a photograph taken afterwards has already lost the number.
5. **Record the temperature.** μ_{eff} depends on T, and a report that quotes a moment without the temperature is incomplete.

요약. 이 저울은 고전적인 Gouy 실험을 뒤집은 것으로, 시료를 매다는 대신 자석을 매달고 시료를 넣었을 때의 반작용을 측정한다. 부피 감수율과 질량 감수율 두 값이 나오며 둘은 충전 밀도로 연결되므로, 시료를 고르게 갈고 균일하게 다져 넣는 것이 질량만큼 중요하다. 핵심은 계산의 함정이다. 페로센은 18전자 저스핀 d^6이라 홀전자가 없고 μ_{eff}는 0이어야 한다. 정상적인 시료라면 χ_M이 반자성 보정값과 비슷한 작은 음수로 나와 보정 후 0이 된다. 그런데 저울이 분해능 한계 때문에 0을 읽으면, 음수인 반자성 보정을 빼면서 오히려 양수가 더해져 0.6–0.8 B.M. 정도의 모멘트가 허공에서 생긴다. 이것은 측정이 감지하지 못한 보정값을 빼서 나온 계산상의 산물이지 홀전자의 증거가 아니다. 측정된 모멘트라고 쓰지 말고 저울 분해능 내에서 반자성이라는 상한으로 서술한다.

Appendix I — The Orbital Diagram, and What It Predicts

I1. Qualitative MO diagram of $Fe(C_5H_5)_2$, staggered D_{5d}



Each Cp^- ring carries five π orbitals holding six electrons; the two rings combine into ten *ligand group orbitals*, in-phase and out-of-phase pairs of each ring orbital, and these are sorted by symmetry against the iron 3*d*, 4*s* and 4*p* set. The strong bonds are the e_{1g} pair, where d_{xz} and d_{yz} overlap with the ring e_1 combinations — that interaction *is* the metal–ring bond. The e_{2g} set interacts only weakly and the a_{1g} (d_{z^2}) hardly at all, so these three orbitals stay largely metal in character and hold the six iron *d* electrons at the top of the occupied manifold. The e_{2u} combination finds no iron orbital of matching symmetry and remains a pure-ligand, non-bonding level, empty. In the eclipsed D_{5h} conformer the labels change (a_1', e_1'', e_2' …) but the picture does not.

I2. Four measurements, one diagram

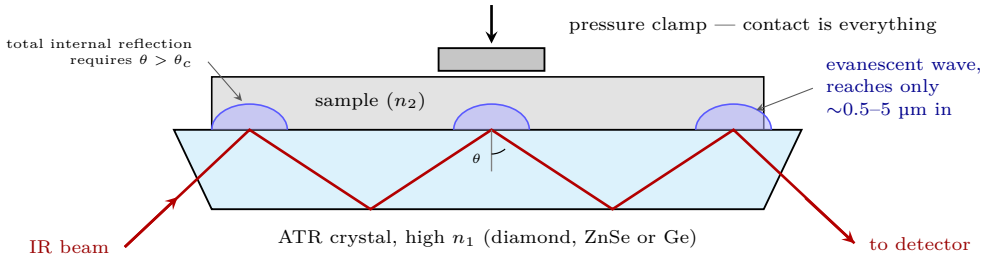
Why 18 electrons is stable	Every one of the eighteen sits in a bonding or non-bonding level and the first empty orbital, e_{1g}^*, is antibonding and well separated. There is no thermodynamic reason for the compound to gain or lose electrons under ordinary conditions, which is why it can be handled, sublimed and stored in air.
Magnetic susceptibility (Appendix H)	The HOMO region is $e_{2g}^4 a_{1g}^2$, a closed shell. **No unpaired electron, so the sample must be diamagnetic** and any positive moment you compute is the arithmetic artefact described in Appendix H, not a property of the molecule.
UV–Vis, the weak bands at ∼440 and ∼325 nm	Both are excitations from the metal-based a_{1g} / e_{2g} pair into the metal-based e_{1g}^*: $d \rightarrow d$ transitions, Laporte-forbidden in a centrosymmetric molecule, hence ε of order 10^2 rather than 10^4. The intense charge-transfer bands lie further into the ultraviolet, below the 300 nm edge of the scan.
ESI-MS gives m/z 186, the ferrocenium cation	Positive-mode electrospray oxidises the analyte at the emitter. Removing one electron from the non-bonding a_{1g} HOMO costs little and leaves the bonding e_{1g} manifold untouched, so $Fe(C_5H_5)_2^+$ is a well-behaved 17-electron ion that survives to the detector intact. This is the same one-electron couple that makes ferrocene a standard redox reference.

The ordering of a_{1g} and e_{2g} at the top of the occupied set is close and has been debated; the assignment shown, with a_{1g} as HOMO, matches the DFT ground-state configuration quoted in the lecture notes and photoelectron data. Nothing in this experiment depends on which of the two is uppermost.

요약. 두 Cp^- 고리의 π 궤도 열 개가 대칭에 따라 철의 3*d*, 4*s*, 4*p*와 짝을 이룬다. 진짜 결합은 d_{xz}, d_{yz}가 고리의 e_1 조합과 겹치는 e_{1g} 쌍이고, e_{2g}는 약하게만, $a_{1g}(d_{z^2})$는 거의 상호작용하지 않아 이 셋이 철의 *d* 전자 여섯을 담은 채 점유 준위의 맨 위에 놓인다. 전자 열여덟 개가 모두 결합성 또는 비결합성 준위에 들어가고 첫 빈 궤도 e_{1g}^*는 반결합성이므로 공기 중에서도 안정하다. 이 그림 하나가 이번 세션의 측정 셋을 설명한다. HOMO 부근이 닫힌껍질이라 반자성이어야 하고, 약한 UV--Vis 흡수 두 개는 금속 기반 a_{1g}/e_{2g}에서 금속 기반 e_{1g}^*로의 Laporte 금지 *d*--*d* 전이이며, ESI-MS의 m/z 186은 비결합성 HOMO에서 전자 하나를 떼어낸 17전자 양이온이 결합 골격을 유지한 채 검출된 것이다.

Appendix C — ATR-IR: How It Works and What It Changes

C1. Total internal reflection and the evanescent wave



The beam is not passed *through* the sample. It is trapped inside a crystal of high refractive index by total internal reflection, which requires the angle of incidence to exceed the critical angle, $\sin\theta_c = n_2/n_1$. At each bounce a standing field — the **evanescent wave** — leaks a short distance into whatever is pressed against the crystal and is attenuated wherever the sample absorbs. The penetration depth is

$$d_p = \frac{\lambda}{2\pi n_1 \sqrt{\sin^2\theta - (n_2/n_1)^2}},$$

so the beam samples only the first few micrometres, and d_p **grows with wavelength**. Low-wavenumber bands are therefore probed more deeply and appear relatively stronger than in a transmission spectrum — an ATR spectrum and a library transmission spectrum of the same compound are not superimposable without correction.

C2. ATR against KBr disc — choose, then record which you chose

	ATR	**KBr disc**
Preparation	None. Place the dry solid on the crystal and clamp.	Grind with dry KBr, press a pellet.
Sample state	Native solid, undiluted.	Diluted in a salt matrix.
Atmosphere	Open bench — CO_2 near 2360 cm^{-1} and water vapour appear.	Enclosed compartment, can be purged.
Intensities	Low-wavenumber bands enhanced ($d_p \propto \lambda$).	True transmission; comparable with libraries.
Main risk	Poor contact gives a weak spectrum; crystal must be cleaned between samples.	KBr is hygroscopic — a broad O–H band near 3400 cm^{-1} is contamination, not sample.
Suits	Thick, strongly absorbing or moisture-sensitive solids.	Reference-quality spectra for comparison.

Working practice. Record a fresh background immediately before the sample, which suppresses most of the atmospheric CO_2 and water. Tighten the clamp until the signal stops improving — with a powder, most weak ATR spectra are contact failures rather than sample problems. Clean the crystal with a solvent that actually dissolves the residue: for the ruthenium complex that means **DCM**, not acetone. Run the sample *dry*; do not dissolve it, since the complex degrades in solution.

C3. What the 500–4000 cm^{-1} window can and cannot tell you

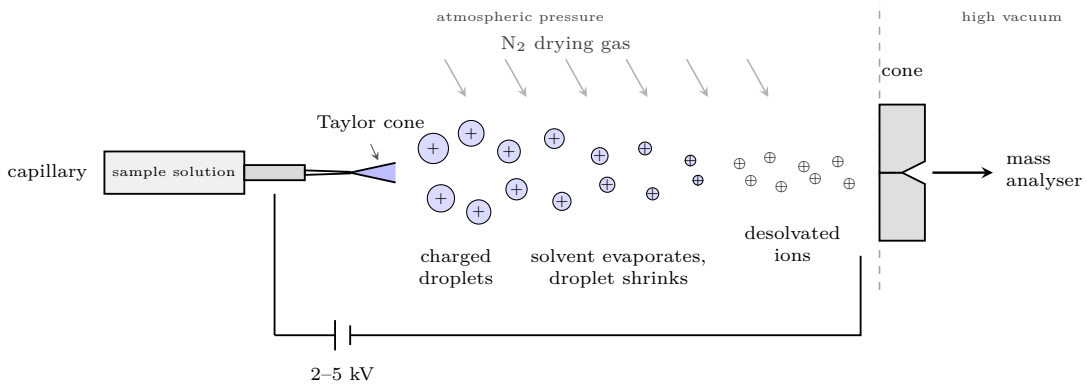
Region / cm^{-1}	**Origin**	**What it tells you**
~3050	Aromatic C–H stretch of the phenyl rings	Phosphine present
~1480, ~1435	Phenyl ring C=C; the lower band is P-phenyl in character	Phosphine present; small shifts on coordination
~1190	P=O stretch of $OPPh_3$	**Should be absent.** Its appearance means the phosphine oxide by-product was not washed out
~1090	X-sensitive phosphine mode	Phosphine present
~745, ~693	Out-of-plane C–H and ring deformation of a monosubstituted ring	Confirms the phenyl substitution pattern
2360	Atmospheric CO_2	**Not your sample.** An open-air ATR artefact

The limitation worth stating in the report. Every band above belongs to the *ligand*. The metal–ligand stretches — Ru–Cl near 300 cm^{-1} and Ru–P lower still — lie below the 500 cm^{-1} cut-off of this measurement, so the spectrum establishes that triphenylphosphine is present and that phosphine oxide is not, but it does not by itself prove that the phosphine is coordinated to ruthenium. That evidence comes from ESI-MS and from the isotope envelope. Treating the IR as confirmation of the *complex* rather than of the *ligand* is the most common overstatement in this report.

요약. ATR은 빛을 시료에 통과시키지 않는다. 굴절률이 높은 결정 안에서 전반사시키고, 반사할 때마다 결정 밖으로 스며 나오는 evanescent wave가 시료 표면 수 마이크로미터만 훑는다. 침투 깊이가 파장에 비례하므로 저파수 흡수가 투과 스펙트럼보다 상대적으로 강하게 나타나며, 라이브러리 스펙트럼과 그대로 겹치지 않는다. 개방된 상태로 측정하므로 2360 cm^{-1} 부근 CO_2는 시료가 아니라 대기이며, 측정 직전에 background를 새로 잡으면 상당 부분 상쇄된다. 스펙트럼이 약하게 나오면 대개 시료 문제가 아니라 결정과의 접촉 불량이므로 클램프를 더 조인다. 마지막으로, 이 창에서 보이는 봉우리는 전부 리간드의 것이다. Ru--Cl는 300 cm^{-1} 부근으로 측정 범위 아래에 있으므로, IR은 포스핀의 존재와 산화물의 부재를 말해 줄 뿐 배위 자체를 증명하지는 않는다.

Appendix B — How ESI-MS Works, and Why It Is Used Here

B1. From solution to gas-phase ions



The solution is pushed through a fine capillary held at 2–5 kV relative to the counter electrode. The field distorts the meniscus into a **Taylor cone** and pulls off a spray of droplets carrying excess charge. A stream of warm N_2 evaporates the solvent, so each droplet shrinks at constant charge until electrostatic repulsion overcomes surface tension and it breaks up — the **Coulombic fission** limit. Repeating this leaves bare, solvent-free ions that pass through the cone into the vacuum stage and on to the analyser.

B2. Why this matters for the interpretation

ESI IS A SOFT TECHNIQUE — THE CONSEQUENCE FOR YOUR SPECTRUM

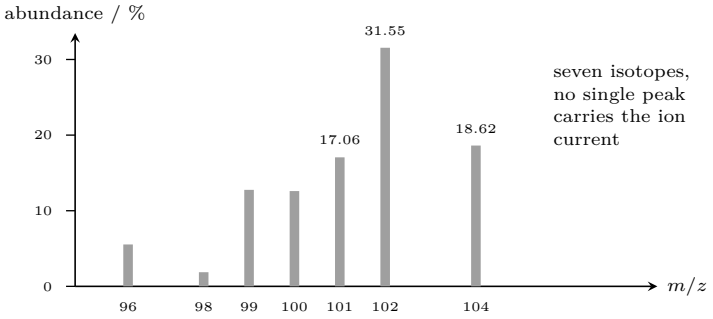
Electron ionization fires high-energy electrons at a *vapourised* sample and shatters it into a reproducible fragment pattern. ESI does none of this: there are no high-energy electrons and no energetic collisions, and the sample is never vapourised. Two things follow.

First, an involatile, thermally fragile organometallic such as $RuCl_2(PPh_3)_3$ can be analysed at all, which EI would not permit.

Second, and more useful, **any ligand loss you observe is chemistry, not instrumentation.** If the base peak corresponds to $[RuCl(PPh_3)_2]^+$ rather than the intact molecule, that reports on the lability of the apical phosphine in solution — the very same dissociation that opens a coordination site in the catalytic cycle. Under EI the same peak would carry no such meaning.

In positive-ion mode the emitter also *oxidises* the analyte. For an easily oxidised species such as ferrocene this gives the radical cation $Fc^{\bullet+}$ directly, rather than a protonated adduct.

B3. Ruthenium isotope pattern — what the cluster should look like



Ruthenium has seven stable isotopes, and chlorine adds $^{35}Cl : {}^{37}Cl = 100 : 32.5$ on top. A ruthenium chloride complex therefore produces a broad envelope of roughly a dozen peaks at unit spacing rather than one line, and **reproducing that envelope is itself evidence that the intended compound was formed**. This is the opposite of organic mass spectrometry, where C and H give weak $M{+}1$ and $M{+}2$ satellites that are easy to ignore. Predicted patterns can be generated with a free isotope distribution calculator; expect some deviation from the measured one.

One artefact to rule out. Ions formed by unimolecular processes scale with source pressure, but bimolecular association scales with its *square*. An overconcentrated sample can therefore generate dimer ions *heavier* than the molecular ion and bury it. If a peak appears at roughly twice the expected mass, dilute the sample and re-run rather than assigning it.

요약. ESI는 고에너지 전자나 충돌을 쓰지 않는 연성 이온화이므로, 스펙트럼에서 관찰되는 리간드 손실은 장비가 만든 조각이 아니라 용액에서 실제로 일어나는 화학이다. 루테늄은 안정 동위원소가 일곱 개여서 한 화합물이 십여 개의 봉우리 묶음으로 나타나며, 이 묶음의 모양이 재현되는지가 목표 화합물이 생성되었다는 증거가 된다. 시료가 너무 진하면 이분자 회합으로 분자이온보다 무거운 이온이 생겨 해석을 방해하므로, 예상보다 두 배가량 무거운 봉우리가 보이면 희석해 다시 측정한다.